Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/03/2017
PI Name:	Sekulic, Dusan D.Sc.		
Project Title:	BRAzing of Aluminum Alloys IN Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sekulicd@engr.uky.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	859-257-2972
Organization Name:	University of Kentucky		
PI Address 1:	CRMS Building, South Limestone		
PI Address 2:	Mechanical Engineering		
PI Web Page:			
City:	Lexington	**State:**	KY
Zip Code:	40506	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/04/2016	**End Date:**	11/03/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
Contact Email:	jan.r.rogers@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2017: No changes		
COI Name (Institution):	Fransaer, Jan Ph.D. (Co-PI/ University of Leuven (KU Leuven), Belgium) Krivilev, Mikhail Ph.D. (Co-PI/ UDMURT STATE UNIVERSITY) Seveno, David Ph.D. (Co-PI/ University of Leuven (KU Leuven), Belgium) Mesarovic, Sinisa Ph.D. (Washington State University)		
Grant/Contract No.:	NNX17AB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to investigate the techniques of brazing (bonding by molten alloy) in microgravity and vacuum condition for space construction and repair of spacecrafts damaged by micrometeorites and orbital debris. A series of space (International Space Station -- ISS) and terrestrial experiments, integrated with modeling and analysis are proposed to answer the practical and fundamental questions: (1) How can we take advantage of the absence of gravity and dominance of capillary forces in transporting the molten braze? (2) What composition of the brazing alloy and the brazing technique will enable strong and efficient bonding of structural elements and repair? (3) Can this brazing be done using fluxless cladding? and (4) Can these brazing clads be synthesized by electrodeposition? Specifically: A comparison between flux and fluxless brazing techniques under space and terrestrial conditions will be performed. The phenomena to be studied under these conditions include molten metal kinetics, wetting and spreading of liquid clad/filler over topographically modified surfaces. The impact of residual oxygen and vacuum conditions outboard of the ISS will be analyzed. The applied aspects include direct verification if brazing of Al alloys is transferable to orbit for rapid repair of the pressure wall. For the first time, brazing in space will be processed using the patented fluxless Trillium technology. Further development includes design and testing of a space brazing furnace for brazing Al truss assemblies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of the proposed research on the life on Earth manifests itself through multifaceted benefits. These include: (i) uncovering so-far hidden physical/chemical/metallurgical aspects of metal bonding by brazing (extensively used in manufacturing) through uncoupling of the gravity impact, hence transforming the art of this technology to a predictive science for terrestrial applications, (ii) exploring a possibility to use novel materials as bonding alloys less sensitive on terrestrial hindrances (e.g., oxidation) in designing bonded assemblies, (iii) featuring sustainable resources use and less energy invested due to an expected broadening the range of applicability of bonding under adverse conditions, (iv) developing visualization of the brazing process in situ and in real time under vacuum and controlled atmosphere, (v) developing the kinetics of wetting theory by implementing phase field techniques that can be used to model capillary phenomena not only for liquid metals but for other material systems in both micro gravity and gravity (on Earth) conditions.
Task Progress:	The overall project work includes tasks distributed among the teams at the University of Kentucky and Washington State University, funded through NASA, with additional planned collaboration with European Space Agency, Belgian Federal Science Policy Office's (ESA, BELSPO) sponsored University of Leuven's team, and Roscosmos' sponsored Udmurt State University. The tasks planned for the first year to be executed by US based teams have involved both experimental and theoretical (numerical modeling development) segments. The participation of Russian and Belgian sides has not yet been funded. Still, an intense communication between all four (two in US, one in Belgium, and one in Russia) teams has been established. Two modes of interactions have been instituted: (i) Skype conferencing, and (ii) meetings of teams' members at either US, EU, or Russian participants' laboratories. Regular Skype meetings have been established. The first joint meeting with physical presence has been organized In Belgium between July 7th and 10th, 2017. Three days-long working sessions have involved all four teams' leaders. Detailed discussion of the Science Requirements Document (SRD) for SCR (Science Concept Review) resulted in a preparation of the first draft SRD document. The representative of the BELSPO participated as well. It should be noted that the activities specified to be executed by the Belgian team (consideration of the alternatives for brazing material synthesis) have not been funded in the Year 1. The new material synthesis (electrodeposition) should be postponed till funding is provided. The Russian team expects also funding in Year 2018 but has been actively participating. The experimental work in Year 1 has been conducted in the Brazing Laboratory of the Institute for Sustainable Manufacturing at the University of Kentucky (the tasks involving experiments on Earth). The work is executed as planned. The experiments utilized the hot stage microscopy, involving samples made of Al alloys with the Al-Si- KFAl alloy in the "hole-in-plate," "pin-in-hole," and "wedge-tee" sample configurations. These experiments have involved uncovering the conditions under which a hole on a flat surface of a certain diameter (in a sub-millimeter range to simulate the hypothesized micro-meteorite material penetration) would be filled under terrestrial conditions, driven only by surface tension. Simultaneously, the preliminary testing of the feasibility of using the existing "Solidification Using a Baffle in Sealed Ampoules" (SUBSA) furnace at the ISS has been performed at the Marshall Space Flight Center, Huntsville, AL. The preliminary results of the heating-cooling cycle between room temperature and the 873 K level indicate a heating rate of up to 10 K/min with multiple Al coupons within the central segment of the hot zone holding a set of experimental coupons. The hot stage microscopy brazing tests have been executed under controlled atmosphere and gravity conditions both present. It is established that a threshold size of the hole diameter exists, with the hole opening non-filled under all considered test conditions if the diameter is larger than the threshold value. A major hypothesis formulated a priori is that this threshold would not exist under microgravity conditions (that is, the planned ISS experiments would verify that hypothesis.) All tests in real time and in situ performed in the hot stage microscopy system have been visualized so a detailed study of the kinetics of the spreading will be possible to perform. An extensive data base of spreading vs. time and topography/material systems selection has been formed with data so far collected. A definite proof of an intense generation of a gas phase during melting of the braze alloy is obtained. The buoyancy effect drives the gas bubbles to the liquid phase surface under terrestrial conditions. Hence, under micro-gravity, an absence of a significant buoyancy would lead to a trapping of the stream of bubbles within the liquid braze filler, thus forming the porosity due to in situ trapped bubbles. This is a significant difference between the terrestrial and ISS conditions to be considered. The Year 1 theoretical work has been focused on: (1) Finite element implementation of the phase field model into the software FreeFEM, (2) Bubble separation problem in microgravity, and, (3) Analysis of equilibrium configurations in the hole filling experiments. Mathematical formulation required for the FreeFEM consists of the weak form of the time discretized governing equations and the derivation of the Jacobian required for Newton's iterations for the nonlinear problem. Analytical work

	is completed and numerical implementation is in progress. The bubbles formed as the result of chemical reaction of the flux with the substrate. Under Earth's gravity, they rise under the buoyancy force and leave the brazing liquid domain resulting in string solid bonds upon solidification of the brazing alloy. The main conclusion of our analysis is that bubble separation from the solid substrate under microgravity (one millionth of the Earth's gravity), requires thousand times larger bubble that on Earth. The experiments with cylindrical hole in a plate and brazing alloy on the top surface of the plate, indicate that hole filling by melted braze depends on the geometry and available liquid volume. The existence of the equilibrium solution which keeps all the liquid on the top surface means that hole filling may not occur. Its stability guarantees that hole is not filled. We have formulated the mathematical problem for equilibrium configurations and are currently performing numerical analysis.
Bibliography Type:	Description: (Last Updated: 12/06/2025)
Awards	Sekulic DP, Yu C-N, Wu Y. (Dusan P. Sekulic, Cheng-Nien Yu, Yangyang Wu) "Invited Keynote Speech at the International forum on Green Material and Processing Technology for Circular Economy, Beijing, China, June 22-24, 2017. Topic: Flux on or flux within! Aluminum brazing advances." Jun-2017
Books/Book Chapters	Bar-Cohen Y, Sekulic DP, Pan R, Babu SS, Hu A, Bao X, Badescu M, Le HJ, Sherrit S. "Metal joining techniques using brazing." in "Advances in manufacturing and processing of materials and structures." Ed. Y. Bar-Cohen. Boca Raton, FL : CRC Press, Taylor & Francis Group, LLC. (in press as of September 2017; expected publication March 2018)., Sep-2017
Books/Book Chapters	Pan R, Sekulic DP. "Diffusion Bonding." in "Advances in manufacturing and processing of materials and structures." Ed. Y. Bar-Cohen. Boca Raton, FL : CRC Press / Taylor & Francis Group, LLC. (in press as of September 2017; expected publication March 2018)., Sep-2017
Significant Media Coverage	Camenisch A, Hahn K, Holloway L. (Editors) "Research project 'Brazing Aluminum Alloys in Space' sponsored by NASA was highlighted and given an introduction in the section of 'In the Spotlight.' The P.I. of this project, Professor Sekulic, is acknowledged." Kentucky Engineering Journal, published by the College of Engineering, University of Kentucky. Issue: spring, page 28, 2017., Feb-2017
Significant Media Coverage	Harder W. "News titled 'UK Professor Leads International Effort to Study Liquid Metal in Space,' about Prof Sekulic's NASA research." UKNow, University of Kentucky News, published online, Feb. 13, 2017. https://uknow.uky.edu/research/uk-professor-leads-international-effort-study-liquid-metal-space , Feb-2017